CM3604 Deep Learning – Part 3: Error Analysis of the Entailment Fact-Checker

1. System and setup

The system scores each fact against Wikipedia passages with DeBERTa-v3-base (MNLI/FEVER/ANLI). Premises are windows of 3 consecutive sentences prefixed with the page title. They are pruned to those sharing at least 40% of the fact's (stemmed, stop-word-free) words, keeping the best 8. The fact is predicted *Supported* if the maximum entailment probability over premises is at least 0.4. This gives **86.4% accuracy (191/221)**: 10 false positives (FP) and 20 false negatives (FN). Sentence-level premises without the title scored 81.0%, and word-overlap-only scored 80.5%. I examined all 10 FPs and the first 10 FNs (FN1–FN10). The remaining 10 FNs are used only as a check on the categories (§3).

2. Fine-grained error categories

FP-A: Plausible-but-unstated inference. The premise supports a related, weaker or more specific claim (synonym, hypernym, location, or world knowledge), and the model treats the fact as entailed although the gold label is NS/IR.

FP-B: Entity mention mistaken for the relation. The fact's entity appears in the premise, but in a different role (e.g. as the source of a degree, or as a broadcaster named in passing). High lexical overlap makes the model entail a relation (*worked at*, *is an X*) the text never states.

FN-A: Composite or evaluative claims. The fact bundles several pieces of information or is vague/evaluative ("best known for", "first handheld…", "popular", "numerous"). No single premise states it, so the model answers *neutral*.

FN-B: Value or reference conflict. The premise contains a close but different value (date, duration), or the fact has an unresolved pronoun, so the model answers *contradiction* even though the gold label is S.

FN-C: Evidence not surfaced. The supporting text is not in the best-scoring premise, either because word-overlap pruning removed it (the fact uses different words from the passage) or because the passage says it differently.

3. Aggregate statistics

Category	Count (examined 10 + 10)	Examples	Count over all 20 FNs
FP-A Plausible-but-unstated inference	7 / 10 FP	FP1, 2, 3, 5, 8, 9, 10	–

Category	Count (examined 10 + 10)	Examples	Count over all 20 FNs
FP-B Entity mention, not relation	3 / 10 FP	FP4, 6, 7	–
FN-A Composite/evaluative claim	3 / 10 FN	FN1, 2, 3	10 / 20
FN-B Value/reference conflict	3 / 10 FN	FN4, 5, 6	4 / 20
FN-C Evidence not surfaced	4 / 10 FN	FN7, 8, 9, 10	6 / 20

Errors are concentrated in few documents. 12 of 20 FNs come from two biographies (Ji Sung and Jonathan Haagensen), and the generic "X is popular / has starred in numerous…" facts alone give 7 FNs. Pruning directly caused 3 of the 20 FNs (FN9, FN11 and FN12 have no premise left), but it cut the number of premises scored from 5,261 single sentences to 975 windows at no accuracy cost.

4. Detailed examples

Example 1 (FP-A). *Fact:* "Nisar Ahmad Kakru was Kashmiri." *Premise (excerpt):* "…(born 26 October 1949 Baramulla, Kashmir, India) was the Chief Justice of the High Court of Andhra Pradesh…" *Gold:* NS. *Predicted:* S (P(entail) = 0.99). *Type:* FP-A. Being born in Kashmir makes "Kashmiri" likely, but it is not stated, and the person served in Andhra Pradesh. The model accepts a plausible world-knowledge leap, whereas the annotators required explicit support.

Example 2 (FP-B). *Fact:* "He has worked at Mayo Clinic." *Premise:* "Oliphant has a Ph.D. in Biomedical Engineering from the Mayo Clinic and B.S. and M.S. degrees … from Brigham Young University. Oliphant was an Assistant Professor … at Brigham Young University…" *Gold:* NS. *Predicted:* S (0.85). *Type:* FP-B. The Mayo Clinic is mentioned only as the place of a degree, and all employment in the text is at BYU. The model matches the entity and a nearby "he has … Mayo Clinic" pattern rather than checking the relation.

Example 3 (FN-B). *Fact:* "He served for many years as a pastor." *Premise:* "Between 1986 and 1989, Malcolm a pastor of the Jerusalem Christian Assembly. … Malcolm served from 1991 – 2000 as Chairman of 'Christian Action for Israel'…" *Gold:* S. *Predicted:* NS (P(contradiction) = 0.82). *Type:* FN-B. The fact's "many years" and pronoun "He" do not align with the 3-year pastorate in this window. The model reads a mismatch in duration and role as contradiction, and does not combine it with other evidence in the passage.

5. Reflection and improvements

The two error directions have different causes. FPs come from over-trusting lexical and topical closeness, and FNs come from claims that need aggregation, or whose evidence is split across passages or worded differently. Possible improvements: (i) require margin over the neutral/contradiction probabilities and not only a threshold on entailment, since most FPs are very confident (P > 0.85) and would need a stricter, calibrated decision rule; (ii) decompose composite facts such as "best known for…" and aggregate evidence over several premises; (iii) replace overlap-only pruning with a softer retriever (embeddings), to rescue FN-C cases such as "Rio de Janeiro is in Brazil"; (iv) resolve pronouns ("He", "They") with the page title before inference. The dev set is small (221 facts, 30 errors), and all thresholds were tuned on it, so the 86.4% figure is probably optimistic.